

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 917+980

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

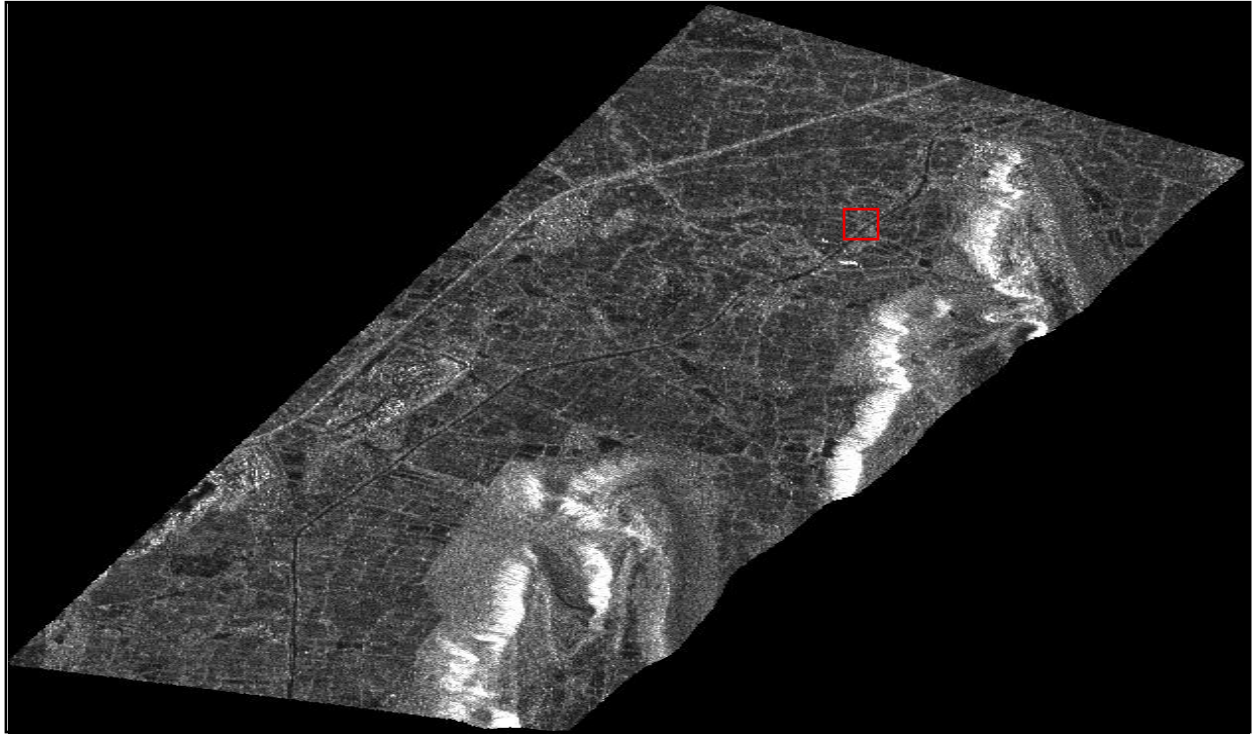
Flight Direction - DESCENDING

Polarization – VV+VH

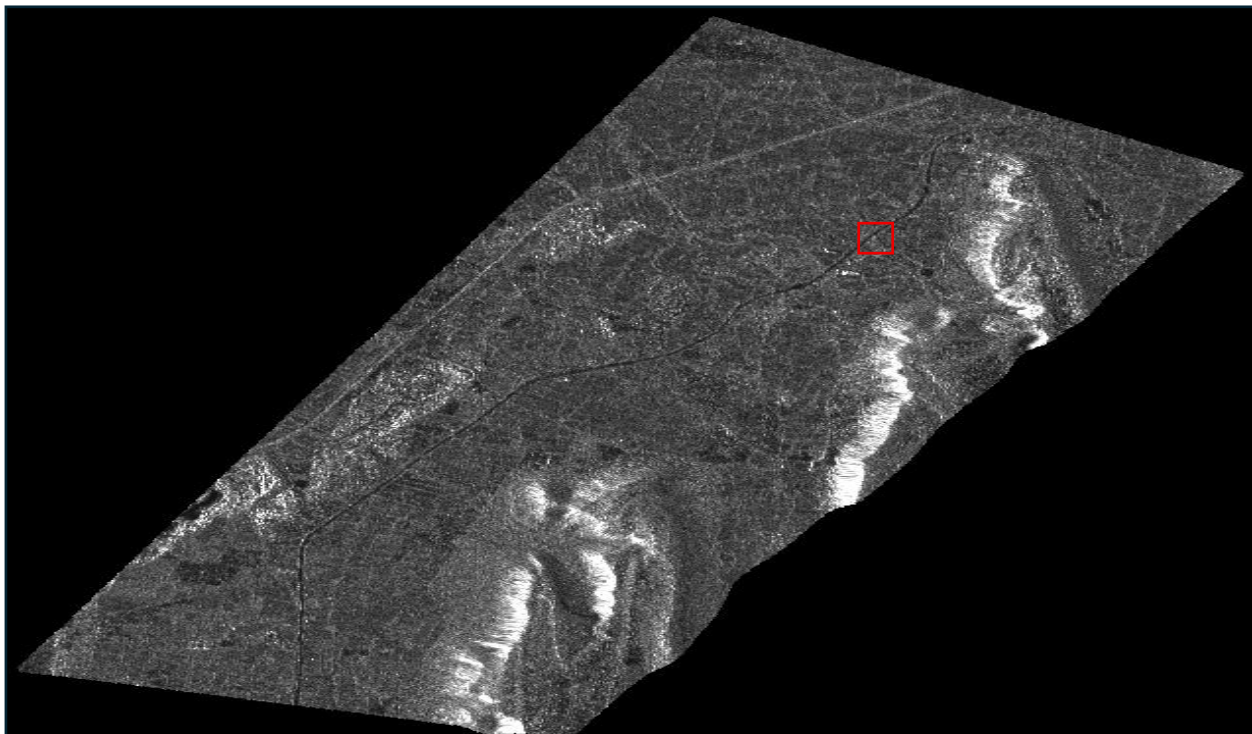
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

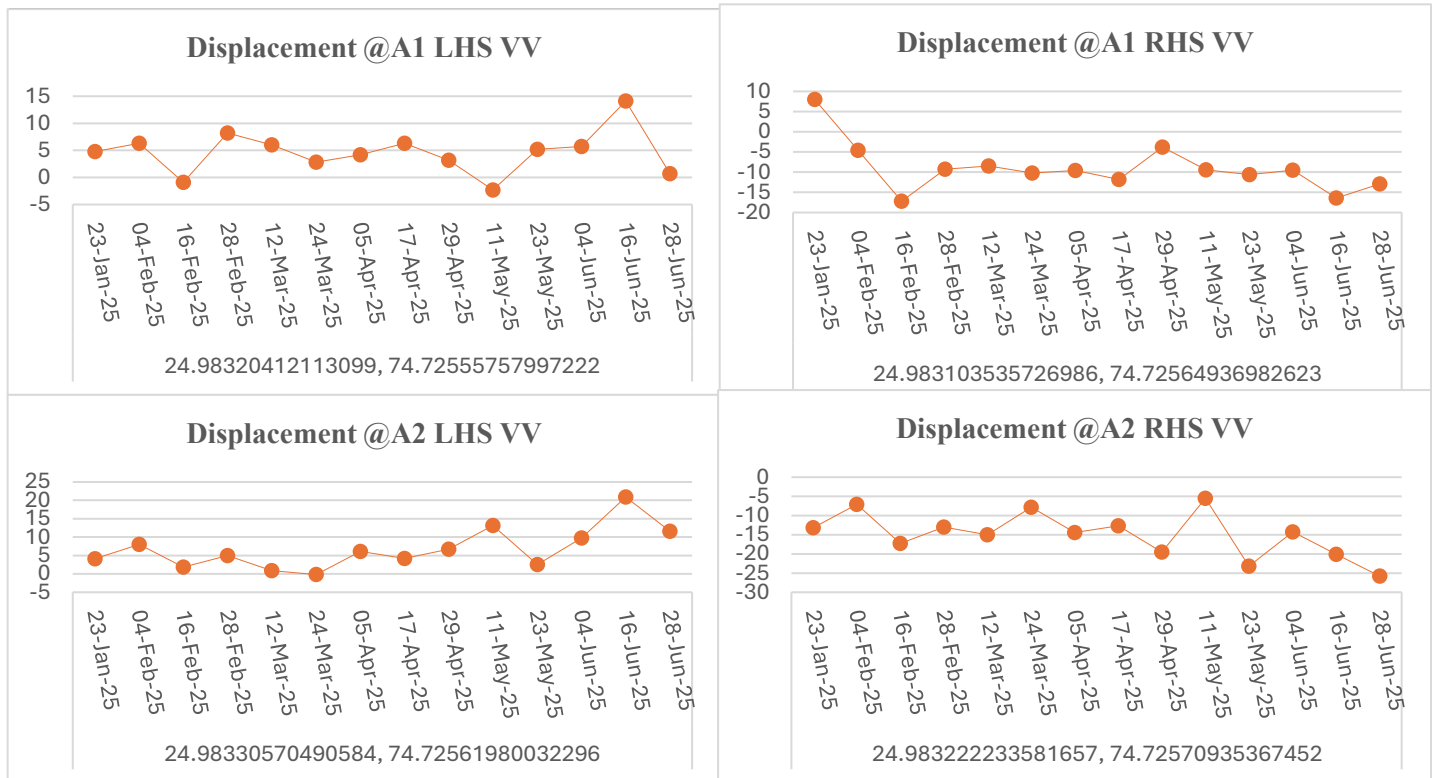
MNB – CH-917+980

Length – 21.97M

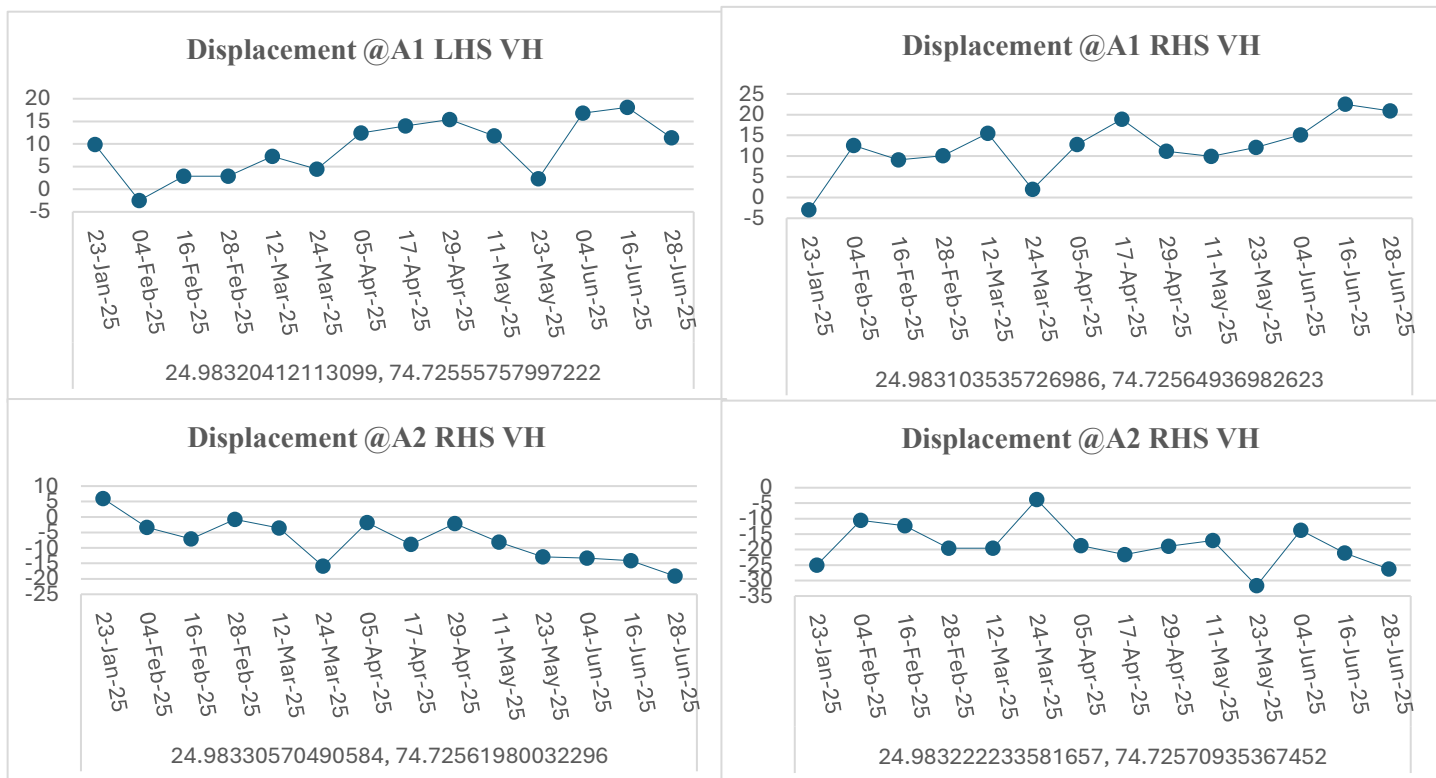


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



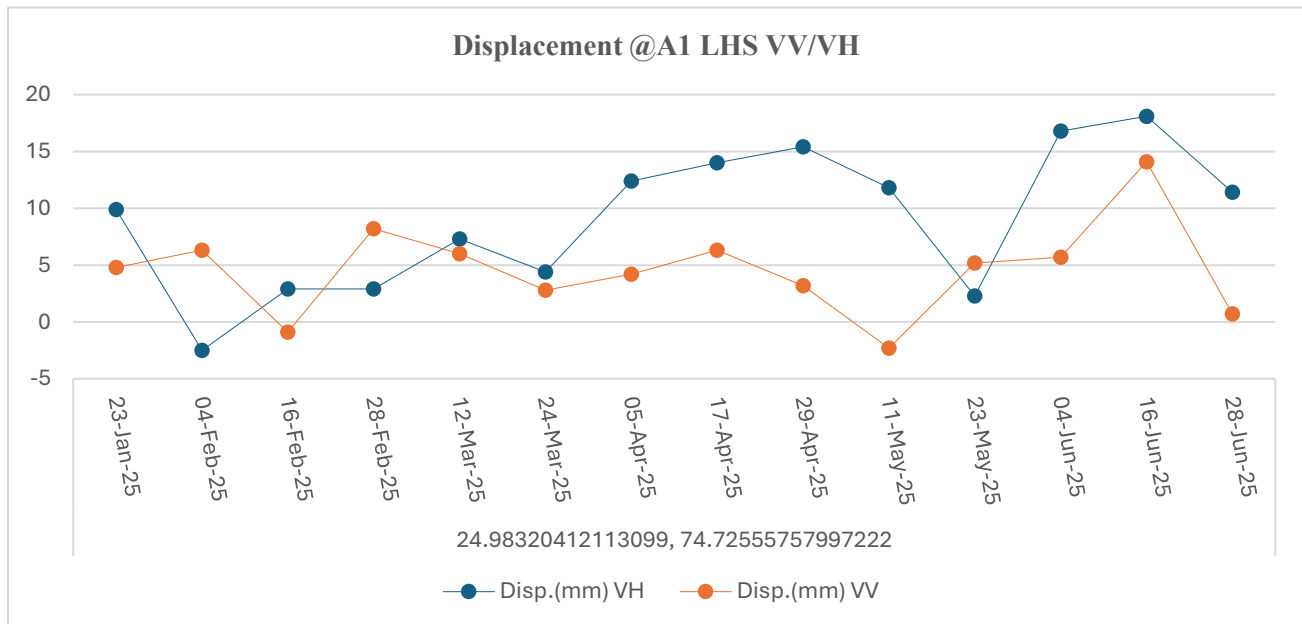
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.98320412113099, 74.72555757997222)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** VV values fluctuate around both positive and negative ranges, approximately between -4 mm and +14 mm, indicating alternating uplift and settlement behaviour.
- **Horizontal Displacement (VH)** VH exhibits predominantly positive values, generally ranging from approximately +2 mm to +18 mm, indicating consistent lateral movement in a single dominant direction at A1 LHS.

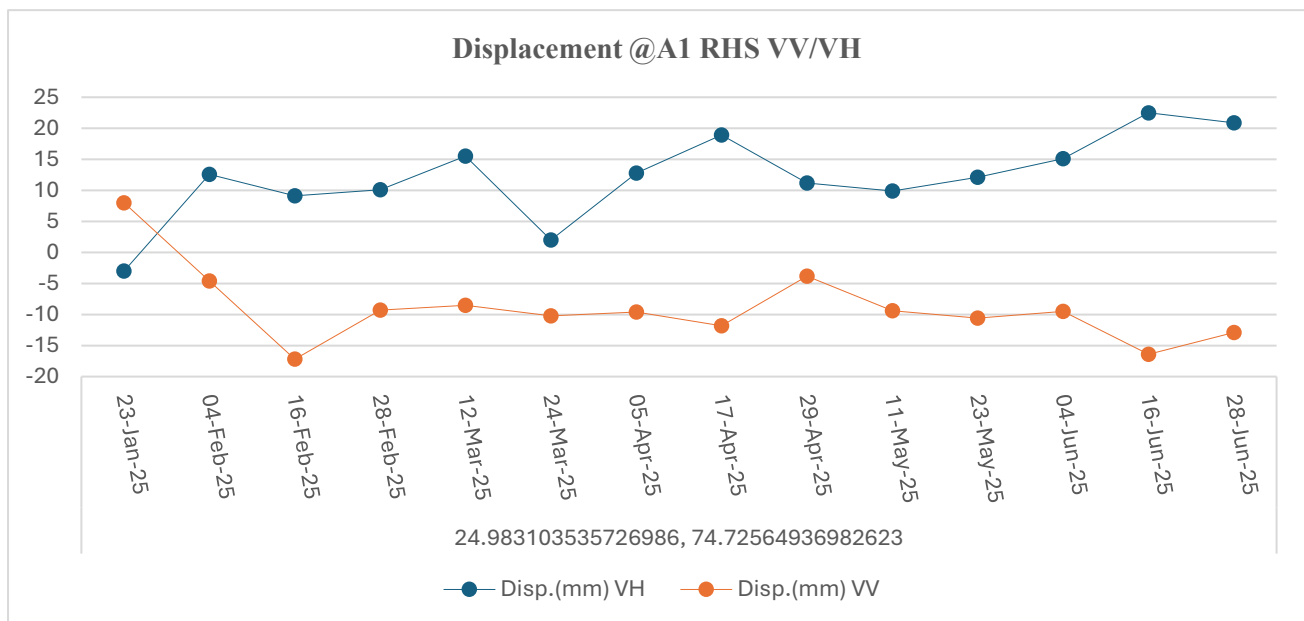
Critical Events & Anomalies:

- **March-April 2025:** A simultaneous increase in VH with moderate VV oscillations indicates dominant lateral structural response, likely influenced by temperature variation or superstructure movement.
- **June 2025:** Both VH and VV exhibit high positive values, indicating a combined lateral shift and vertical uplift, consistent with thermal expansion or transient loading effects.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.983103535726986, 74.72564936982623)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** values are predominantly positive throughout the monitoring period, generally ranging from approximately +2 mm to +23 mm, indicating consistent lateral movement in a single dominant direction at A1 RHS.
- **Vertical displacement (VV)** values are predominantly negative, generally ranging from approximately -5 mm to -18 mm, indicating dominant downward movement or settlement behaviour at A1 RHS.

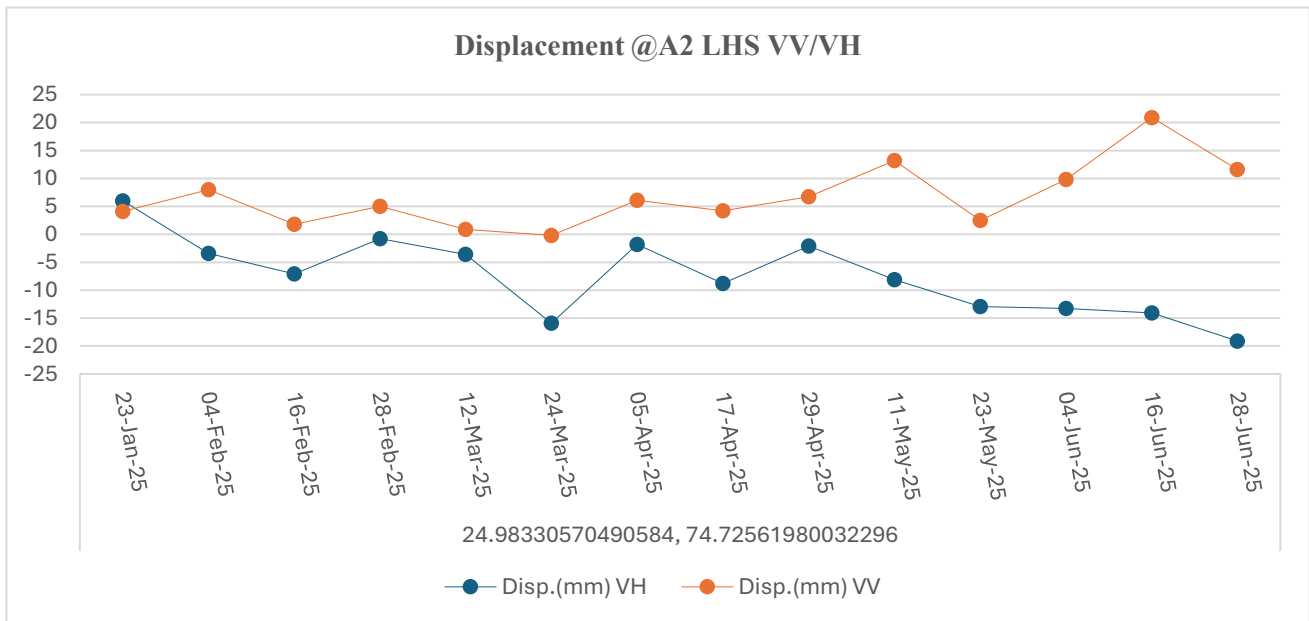
Critical Events & Anomalies:

- **February 2025:** A sharp negative VV excursion combined with increasing VH suggests early-stage foundation or bearing adjustment under operational or environmental loading.
- **June 2025:** Simultaneous maximum VH and deep negative VV indicate a coupled lateral shift with vertical compression, potentially associated with temperature extremes or transient live loading.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.98330570490584, 74.72561980032296)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** values are predominantly negative throughout the monitoring period, generally ranging from approximately -3 mm to -20 mm, indicating consistent lateral movement in a single dominant direction at A2 LHS.
- **Vertical Displacement (VV)** VV values are predominantly positive, generally ranging from approximately +2 mm to +22 mm, indicating overall upward movement or uplift behaviour at A2 LHS.

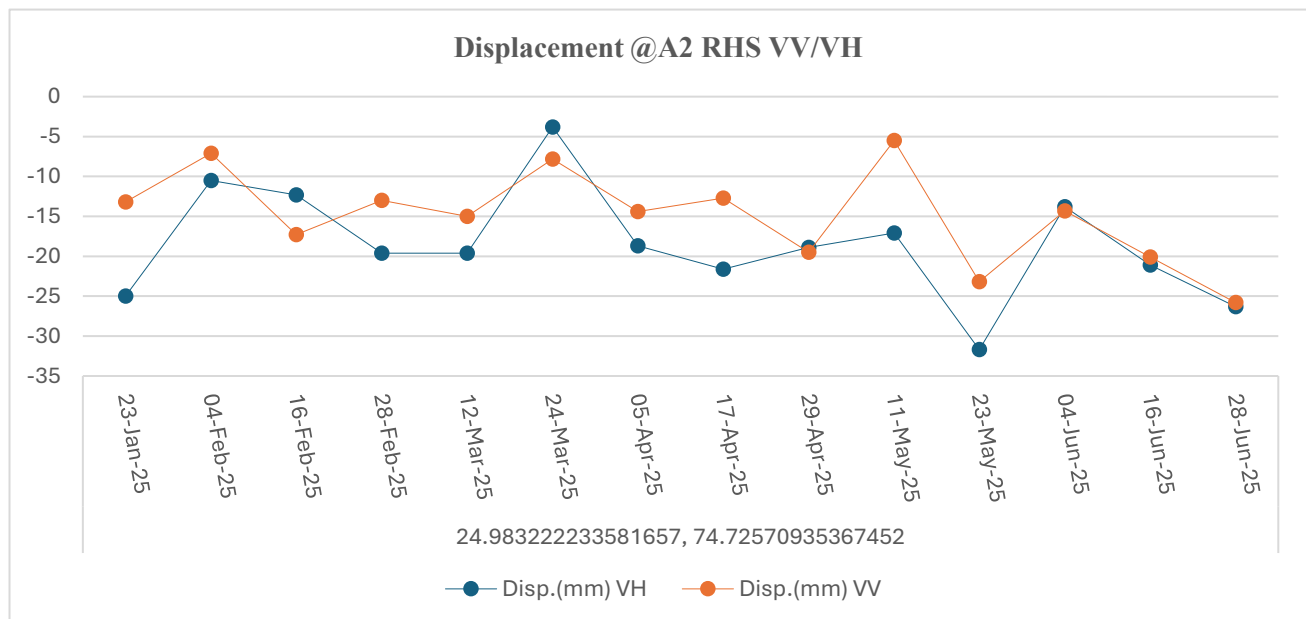
Critical Events & Anomalies:

- **March 2025:** A sharp negative VH excursion combined with moderate VV indicates a localized lateral response, possibly linked to bearing restraint effects or transient loading.
- **April-May 2025:** VH recovery alongside increasing VV suggests combined lateral adjustment and vertical uplift.
- **June 2025:** Simultaneous maximum uplift (VV) and maximum negative VH indicate a coupled rotational or torsional response at A2 LHS.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.983222233581657, 74.72570935367452)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical displacement (VV)** VV values are also predominantly negative, generally ranging from approximately -8 mm to -25 mm, indicating overall downward movement or settlement behaviour at A2 RHS.
- **Horizontal displacement (VH)** values are consistently negative across the entire monitoring period, ranging approximately from -10 mm to -33 mm, indicating persistent lateral movement in a single dominant direction at A2 RHS.

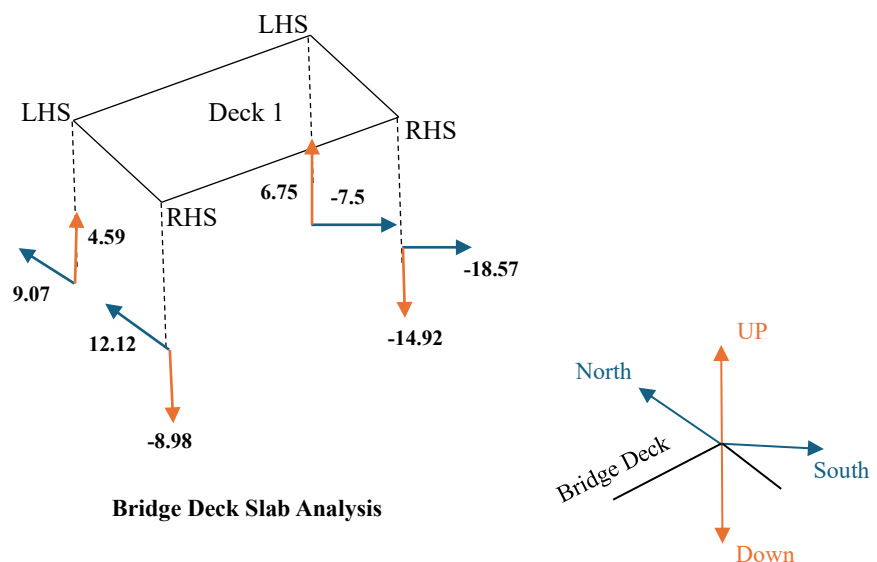
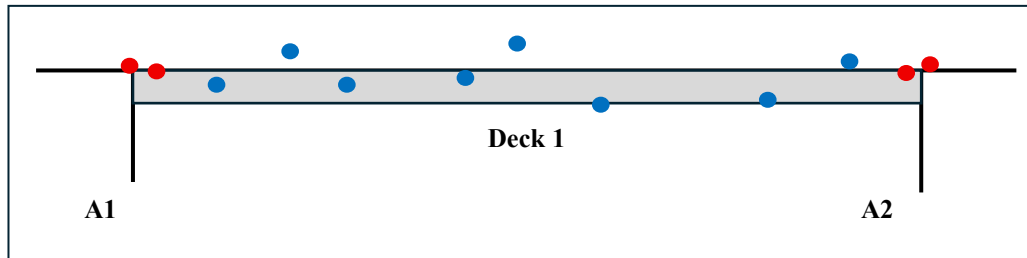
Critical Events & Anomalies:

- **March 2025:** VV briefly reduces in magnitude while VH remains strongly negative, indicating a decoupled response, possibly influenced by temporary load redistribution or thermal effects.
- **June 2025:** Both VH and VV remain strongly negative, suggesting permanent or semi-permanent deformation rather than cyclic movement.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits marked non-uniform vertical displacement, indicating differential behaviour across the deck width and depth. The maximum uplift is observed at the upper central deck region, reaching approximately +6.8 mm, indicating moderate vertical response at the deck crown.

Horizontal Displacement Behaviour:

- Horizontal displacement displays strong directional variability, reflecting asymmetric deformation behaviour. The contrast between LHS and RHS horizontal movements highlights torsional rotation of the deck slab. The LHS region shows substantial lateral movement, with values of approximately +9.1 mm and +12.1 mm, indicating dominant horizontal displacement toward one direction

CONCLUSION:

- The bridge deck slab demonstrates a clearly non-symmetric deformation pattern, characterized by uplift on the upper deck regions and significant settlement at lower RHS locations. The presence of large opposing horizontal displacements on the LHS and RHS confirms torsional behaviour rather than uniform translation. Overall, the displacement pattern suggests uneven stiffness distribution or asymmetric loading effects, warranting continued monitoring and focused assessment of RHS support conditions.